

When the Signal Isn't in the Text: Nine Signals and Debate Fail to Recover PubMedQA maybe

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Abstract

Nine uncertainty signals and compute-matched debate fail to recover PubMedQA maybe; the deliverable is coverage-constrained abstention, not retrieval or multi-agent accuracy. On PQA-L 500, hit@1 is 0.980 [0.968, 0.992] (citation-support 1.000) because gold is the source abstract; BioLinkBERT reaches only 0.726 [0.688, 0.764]. Eight of nine headline AUROC CIs include 0.5 (gpt-5 0.554). On the same 90 IDs, ungated debate is identical to BioLinkBERT (0.656 [0.556, 0.744], 0 discordant pairs) and compute-matched self-consistency ($N=8$, 8.0 LLM calls/case) is 0.522 [0.422, 0.622]. Selective prediction cuts expected cost -16.4% versus always-answer 0.356 at $\geq 50\%$ coverage (AURC 0.311, unanchored vs. a ranker).

Keywords: abstention, selective prediction, uncertainty, PubMedQA, negative result

Data and Code Availability. Official PubMedQA PQA-L (Jin et al. 2019) (public). Per-case signals, bootstrap statistics (statistics.json), and figure scripts ship as anonymized supplemental material. No deanonymizing repository URL at review. Upon acceptance we will release evaluation scripts and reports. IRB. Public research abstracts and official labels only; no patient-identifiable records collected. No live human-subjects study was run. Control C3 uses the official PubMedQA single-annotator labels as a proxy, not a new annotation. No IRB approval is required. Publication Ethics / LLM use. Debate, audit, and self-consistency calls are part of the method; the authors take responsibility for all reported numbers and the bibliography. The manuscript and evaluation scripts were drafted with AI writing/coding assistance; every claim was checked against the released JSON.

1 Introduction

A system that answers a question the evidence cannot support is worse than one that declines. PubMedQA (Jin et al. 2019) operationalises that requirement: given a research question and the abstract that motivated it, the label is yes, no, or maybe — and maybe means inconclusiveness, not a third topic.

Headline accuracies of $\sim 78\text{--}84\%$ are driven by yes/no. Replacing an LLM judge with calibrated BioLinkBERT (Yasunaga et al. 2022) raised label accuracy

from 0.536 to 0.726 [0.688, 0.764] on the same PQA-L 500 run; hit@1 stayed 0.980. That near-ceiling is expected: gold is the source abstract in a 500-abstract corpus, so retrieval contributes nothing; the gap is that the abstract does not decide the label. maybe recall is still only 0.073 (4/55; Figure 1).

We measure nine uncertainty signals, a compute-matched debate, and a fixed DeBERTa-NLI auditor (Abdaljalil et al. 2026). The findings are consistently negative, and that is the result. Across judges and signals, maybe is statistically indistinguishable from chance in the abstract (Figure 2). Three controls localise the cause: (C1) the NLI auditor is at chance (AUROC 0.497); (C2) giving the authors' gold conclusion lifts gpt-5 only to 0.592 (CI still includes 0.5); (C3) the official PubMedQA single annotator reaches 0.60 maybe recall versus ≈ 0 for every model ($p=0.0002$) yet still errs on 40%. Relative to Lionetti et al. (2025) (ML4H 2025 Findings: uncertain labels) and León Tramontini et al. (2026) (ML4H 2025 Proceedings: abstention in trial matching), this is a complementary PubMedQA regime: inconclusiveness is often not in the text, so the practice question is when to refuse.

We therefore treat the deliverable as selective prediction, not recovered maybe accuracy. On the same 90 IDs, ungated debate is identical to the classifier (0.656 [0.556, 0.744], 0 discordant, McNemar $p=1$) — a collapse control, not a second system — and compute-matched SC is 0.522 [0.422, 0.622] at 8 LLM calls each (Section 4). Contributions. (1) A complementary PubMedQA regime vs. Lionetti and Tramontini: inconclusiveness often is not in the text; the practice question is when to refuse. (2) A negative result: nine headline signals, four judges including gpt-5, bootstrap CIs, official single-annotator control C3, and seven-seed routing instability — maybe is not recoverable from the abstract. (3) Coverage-constrained abstention: a 16.4% expected-cost cut versus always-answer 0.356 at $\geq 50\%$ coverage on $n=90$ (AURC 0.311 is unanchored vs. a ranker), with all artefacts released.

2 Related Work

Multi-agent debate was introduced to improve factuality (Du et al. 2024) and adopted in medicine (Tang

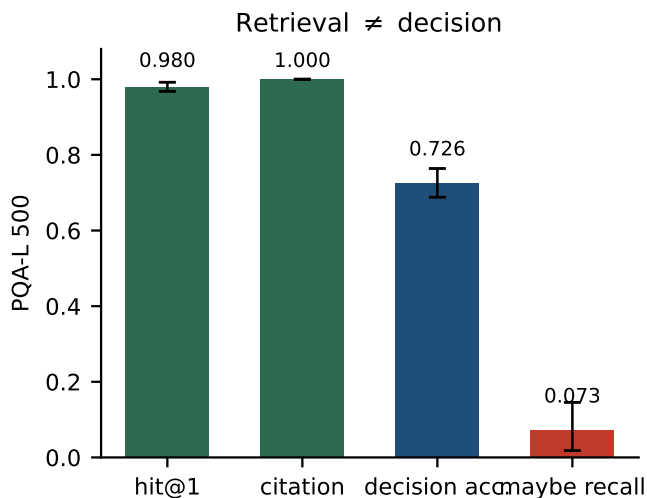


Figure 1: Stage-separated metrics on PQA-L 500 (95% CI). Retrieval (hit@1 0.980, citation 1.000) and BioLinkBERT decision (0.726) do not overlap; maybe recall is 0.073. Source: supplemental statistics.json.

et al. 2024; Kim et al. 2024; Chen et al. 2024), yet compute-matched controls and MedAgentBoard find it rarely beats a strong single model (Wang et al. 2023; Smit et al. 2024; Choi et al. 2025; Zhang et al. 2025; Zhu et al. 2025). We measure the same collapse on PubMedQA maybe (unanimous on 93% of true-maybe cases). Selective prediction (El-Yaniv and Wiener 2010) lets a model abstain; MedQAbstain (Cocchieri et al. 2026) shows LLMs almost never do on medical MCQA, and a fixed NLI auditor (Abdaljalil et al. 2026) is AUROC 0.497 here (C1). Lionetti et al. (2025) (ML4H 2025 Findings) treat clinical labels as uncertain; León Tramoncini et al. (2026) (Proceedings 2025) study abstention in trial matching; SCARE (Lee et al. 2026b) classifies SQL/EHR answerability — complementary PubMedQA work asks whether the abstract is conclusive. Semantic entropy (Kuhn et al. 2023) assumes recoverability from the input; maybe tests the opposite.

3 Method

Three layers share one LLM backend. Debate is a collapse control (identical to BioLinkBERT, 0 discordant, $n=90$), not an architectural contribution; the panel score u -score and a calibrated router remain. An NLI auditor and a gold-conclusion oracle are controls, not accuracy boosters.

Debate. Debate is a collapse control, not an architectural contribution: on $n=90$ ungated debate is identical to BioLinkBERT (0.656 [0.556, 0.744], 59/90, 0 discordant pairs). We run supervisor-free round-robin (round_robin_no_supervisor): round 1 independent; rounds 2–3 sequential with peer context; four personas

including an uncertainty advocate. Each opinion emits a label, evidence_conclusiveness, and a confidence; a two-step rule assigns maybe when Step 1 is inconclusive.

BioLinkBERT gate. A calibrated BioLinkBERT-large classifier is the decision baseline (0.726 on PQA-L 500). A confident (≥ 0.90) hard yes/no is trusted unless the panel is unanimous on the opposite hard label. Among errors on gold maybe, 21/26 mispredictions on balanced90 had confidence ≥ 0.90 , so threshold gating cannot recover abstention. Panel majority without the gate (round-1 vote) is 0.567 [0.467, 0.667] on the same IDs.

u -score and routing. $u = \text{clip}_{[0,1]}(\sum_k w_k f_k)$ over label entropy, maybe and inconclusive fractions, disagreement, flip rate, semantic entropy (Kuhn et al. 2023), a BioLinkBERT-is-maybe bit, and an optional audit score. `calibrate_threshold` sweeps t on a held-out split (default macro- F_1) and routes to maybe iff $u \geq t$. On this run routing is not the primary accuracy: held-out $n=45$ drops ungated 0.644 to 0.622. We report the risk-coverage curve, AUROC, and a cost-sensitive analysis (wrong=1.0, abstain=0.25) with a ≥ 0.5 coverage constraint.

Auditors. The generative auditor decomposes the question into 2–4 checkable conditions (supported/refuted/silent). The external auditor is fixed cross-encoder/nli-deberta-v3-base: high score when the abstract is neutral to both polarities (Abdaljalil et al. 2026).

4 Experimental Setup

Data. Official PubMedQA PQA-L (Jin et al. 2019). Full-scale metrics use PQA-L 500 (276 yes, 169 no, 55 maybe). Signal AUROC uses balanced90 (30 each) so maybe-vs-rest has 30 positives.

Pinned models. Debate and generative audit: qwen2.5:7b (primary), qwen2.5:14b, deepseek-r1:14b. Flagship control: gpt-5 (OpenAI API). Decision classifier: BioLinkBERT-large, pinned checkpoint `pubmedqa_biolinkbert_seed47` (training seed 47). External auditor: cross-encoder/nli-deberta-v3-base. Retrieval: MedCPT (dense) + BM25 + MedCPT cross-encoder reranker. Debate uses $T=0.3$; SC uses $T=0.7$ because sample diversity is the SC mechanism. Frozen abstract for both generative arms. LLM calls (debate, audit, SC) are part of the method; authors take responsibility for the numbers they produce.

Metrics. AUROC (maybe vs rest) with 95% bootstrap CIs (5000 resamples, seed 47). Human comparison: maybe recall, bootstrap difference CI, permutation p . Selective prediction: risk-coverage, AUROC, expected

cost with ≥ 0.5 coverage. Routing stability: seven calibration seeds.

Controls. C1: swap the generative auditor for DeBERTa-NLI. C2: feed the authors’ long_answer instead of the abstract. C3: official PubMedQA single-annotator labels on balanced90 (a proxy; no new human-subjects study).

Self-consistency. The compute-matched control is $N=8$ independent generalist samples, majority vote, frozen abstract, qwen2.5:7b. N equals mean_llm_calls_per_case = 8.0 from the instrumented debate run on the same 90 IDs (both arms: 720 calls, 0 failures). Primary debate metric is ungated bert_gate on all 90, not the routed held-out $n=45$ summary. Debate does not beat BioLinkBERT; SC is worse on the point estimate and is not a significant difference.

5 Results

Retrieval $\neq$ decision. Figure 1 separates the stages on PQA-L 500. Hit@1 is 1 iff the gold source PMID — the PubMedQA source abstract the question was written from — is the top-1 hit in the official 500-abstract corpus; citation-support passes iff every inline $[S\#]$ cites a retrieved document in canonical form (structural validation, not claim-level entailment). Hit@1 = hit@3 = 0.980 [0.968, 0.992]; citation-support is 1.000. That near-ceiling is expected on this construction; the decision is the gap. Headline BioLinkBERT accuracy is 0.726 [0.688, 0.764] on $n=500$ (LLM judge on the same run: 0.536; hit@1 unchanged), not the diagnostic-split 0.656 [0.556, 0.744] on balanced90. The intervals do not overlap retrieval. maybe recall is 0.073 [0.018, 0.145]: only 23/500 predicted maybe against 55 gold (26 collapsed to yes, 25 to no). On balanced90 the same checkpoint has maybe recall 0.133 (4/30). The evidence is found; it does not decide the question.

Compute-matched debate vs. SC, same 90 IDs. Table 1 is the inviolable comparison. Ungated debate (bert_gate, qwen2.5:7b, frozen abstract) is identical to BioLinkBERT: 0.656 [0.556, 0.744] (59/90), 0 discordant pairs, McNemar $p=1$, maybe recall 0.133. We report that collapse, not a second competing system. The panel without the gate — round-1 majority vote, no BioLinkBERT — is 0.567 [0.467, 0.667] (51/90) on the same IDs. Compute-matched SC ($N=8$, 8.0 LLM calls/case, $T=0.7$ vs. debate $T=0.3$) is 0.522 [0.422, 0.622] (47/90); its higher maybe recall 0.567 (17/30) comes from over-predicting maybe (45/90). Matched 8 calls, debate still spends ~ 10.6 k tokens/case versus SC ~ 8.0 k — no accuracy win over BERT on the same 90 IDs. McNemar exact $p=0.088$ (debate vs. SC; 27 vs. 15 discordant) is a point-estimate gap only. Routing is not primary: on the

Table 1: Decision accuracy. Headline BioLinkBERT is 0.726 on PQA-L 500. Apples-to-apples rows share the same 90 IDs, frozen abstract, qwen2.5:7b. Debate ungated is bert_gate — identical to BERT, not a second system. Panel maj. is round-1 majority without BERT. $n=90$ CIs: case bootstrap, 5000 resamples, seed 47. Routed held-out $n=45$ accuracy 0.622 is not primary.[†]

System	Acc.	95% CI	n	Calls
BERT (PQA-L 500)	0.726	[0.688, 0.764]	500	—
BERT (bal. 90)	0.656	[0.556, 0.744]	90	—
Debate ungated	0.656	[0.556, 0.744]	90	8.0
Panel maj. (no BERT)	0.567	[0.467, 0.667]	90	8.0
SC ($N=8$)	0.522	[0.422, 0.622]	90	8.0

[†]Routed debate on held-out 45: 0.622 (drop from ungated 0.644 on those IDs; SC 0.533). McNemar debate vs. BERT: 0 discordant, $p=1$.

Debate vs. SC: $p=0.088$ — point estimate only.

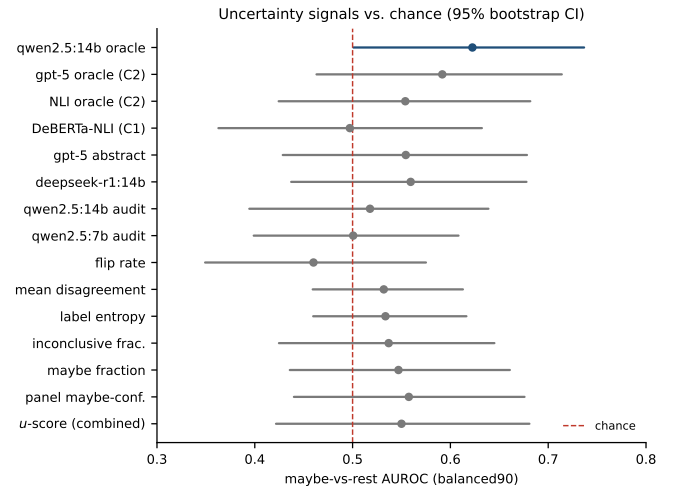


Figure 2: maybe-vs-rest AUROC, balanced90, 5000 bootstrap resamples (seed 47). Grey: CI includes 0.5. Navy: sole interval that excludes chance (qwen2.5:14b gold-conclusion oracle). Eight of nine headline CIs include 0.5. Source: supplemental statistics.json.

held-out 45 it drops ungated 0.644 to 0.622 (SC on those IDs is 0.533). 95% CIs on $n=90$ are case-level bootstrap (5000 resamples, seed 47) from the per-case labels in the two JSON reports.

No uncertainty signal beats chance. Figure 2 collects maybe-vs-rest AUROC on balanced90. Point estimates sit in 0.46–0.62. Nine headline signals are reported (u -score, four generative audits, DeBERTa-NLI, two gold-conclusion oracles, plus the NLI oracle). Eight of nine 95% CIs include 0.5, including gpt-5 (0.554 [0.429, 0.678]) and DeBERTa-NLI (0.497 [0.363, 0.632]). The only interval that clears 0.5 is the qwen2.5:14b gold-conclusion oracle (0.622 [0.501, 0.737]), and it barely does. The gpt-5 oracle is 0.592 [0.463, 0.714] — still chance.

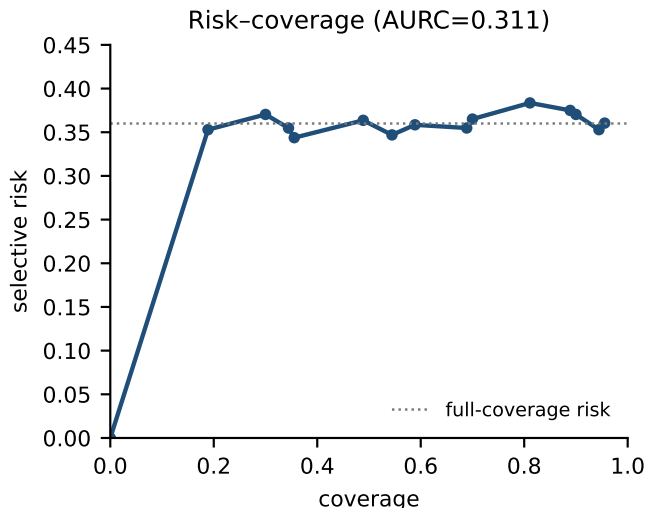


Figure 3: Risk-coverage for the routed u -score on balanced90 ($n=90$). AURC 0.311 (no ranker baseline in the released files). The usable operating point is constrained $\geq 50\%$ coverage (Table 2), not unconstrained abstain-all.

Table 2: Cost-sensitive selective prediction on $n=90$ (wrong=1.0, abstain=0.25). A predicted maybe costs 0.25, so always-answer is 0.356, not the 0-1 error rate 0.344.

Point	Cost	Abstain	n	vs always
Always answer	0.356	0%	90	—
Unconst. opt.	0.250	100%	90	−29.7% (deg.)
$\geq 50\%$ cov.	0.297	48%	90	−16.4%

Official annotator beats models; the residual is partly irreducible. Official single-annotator predictions (C3) on balanced90: maybe recall 0.60 (accuracy 0.800, $\kappa=0.70$) versus ≈ 0.00 for every model signal. Difference 0.60, 95% CI [0.43, 0.77], permutation $p=0.0002$. Giving C3 the gold long_answer lowers overall accuracy 0.800 $\rightarrow$ 0.711 while maybe recall only moves 0.60 $\rightarrow$ 0.63.

Routing is unstable; selective prediction still pays. Seven-seed routing (macro- F_1) yields accuracy 0.587 ± 0.075 and maybe recall 0.257 ± 0.144 (range [0.067, 0.467]). The spread is evidence that the signal is too weak to tune. Nonetheless the risk-coverage AURC is 0.311 (Figure 3). No random or BERT-confidence AURC is in the released analysis files, so 0.311 is unanchored as a ranker comparison; we treat always-answer expected cost 0.356 ($n=90$) as the operating-point anchor. Unconstrained cost optimisation degenerates to abstain-on-everything. The constrained point — answer at least half of cases — cuts expected cost from 0.356 to 0.297 (−16.4%) at 52% coverage (Table 2). Always-answer 0.356 is not $1 - 0.656 = 0.344$: a predicted maybe is charged as abstention (0.25), so 31 errors and 4 correct maybes give $(31+1)/90=0.356$.

6 Discussion

The deciding information for many PubMedQA maybe items is not in the abstract the model reads. The failure is invariant to the answering procedure (nine headline signals, four judges, a dedicated NLI auditor). It is not retrieval (hit@1 0.98 on a source-abstract gold; CIs do not overlap decision accuracy). Injecting the gold conclusion moves the best oracle only to 0.62, and the official single annotator (C3) still errs on 40%. Others improve how a model answers (Abdaljalil et al. 2026; Cocchieri et al. 2026; Wynn et al. 2025); here the problem is upstream.

We do not claim SOTA or that debate improves anything. Ungated debate is identical to BioLinkBERT (0.656 [0.556, 0.744], 0 discordant, McNemar $p=1$). SC is lower at matched compute (0.522 [0.422, 0.622]); McNemar $p=0.088$ is not a significance claim. Routed debate (0.622 on held-out 45) is not primary. NLI AUROC 0.497 is a control failure. Limits: debate/SC were not run on PQA-L 500; one benchmark; contamination is possible; bert_gate is a BERT copy and the panel-only majority is 0.567; C3 is the official single annotator, not a new multi-rater study; $\geq 70B$ open models were not run.

The −16.4% figure is a coverage-constrained cost cut versus always-answer on $n=90$, not a ranker-anchored AURC win and not a bedside claim. SCARE (Lee et al. 2026b) and FHIR-AgentBench (Lee et al. 2026a) ask SQL/EHR answerability; we ask whether the abstract is conclusive. The symposium argument is label versus abstain, not a new architecture. Per-case signals and statistics ship as anonymized supplemental material.

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